

## AI-Powered Recruiter CV Analyzer with Conversational Hiring Assistant

### Problem Statement

In modern recruitment, companies often receive hundreds or even thousands of resumes for a single job opening. Recruiters manually screening these resumes face multiple challenges: large volume of applications, high screening time, human fatigue, bias during shortlisting, and difficulty identifying relevant candidates.

### Proposed Solution

We propose an AI-powered CV analyzer using OCR, resume parsing, Large Language Models (LLMs), and conversational AI. Recruiters upload multiple resumes and a job description. The system reads resumes, extracts structured candidate information, compares each candidate with job requirements, generates match scores, ranks candidates, and provides explainable insights.

### Workflow

Recruiter Uploads CVs + Job Description

- Input Validation Layer
- Resume Parsing / OCR Engine
- Structured Candidate Profiles
- LLM Evaluation Engine
- Candidate Scoring Engine
- Ranking and Shortlisting
- AI Recruiter Chat Assistant
- Final Hiring Insights

### Step 1: Input Layer

Recruiter uploads resumes (PDF/JPG/PNG) and a Job Description containing role, required skills, and preferred skills.

### Step 2: Input Validation

Checks if files are readable, corrupted, or blurry before processing.

### Step 3: Resume Parsing and OCR

PDF resumes use text parsing.

Image resumes use OCR to extract text.

### Step 4: Information Extraction

Extract:

- Candidate Name
- Skills
- Education
- Projects
- Experience
- Certifications
- Portfolio Links

### Step 5: LLM Evaluation Engine

LLM evaluates:

- Skill match
- Project relevance
- Experience quality
- Resume quality
- Missing requirements

### Step 6: Candidate Scoring

Weights:

Skill Match: 40%

Project Relevance: 25%  
Experience: 20%  
Resume Quality: 15%

Final Score =  
 $0.40 \times \text{Skill Match} +$   
 $0.25 \times \text{Project Relevance} +$   
 $0.20 \times \text{Experience} +$   
 $0.15 \times \text{Resume Quality}$

Step 7: Ranking  
Candidates ranked by score for recruiter review.

Step 8: Explainability Layer  
Each score includes strengths and weaknesses for transparency.

Step 9: Conversational Recruiter Chatbot  
Recruiter can ask:  
- Who are top 3 candidates?  
- Why was candidate B ranked lower?  
- Who is best for robotics role?

Tech Stack  
Frontend: Streamlit / React  
Backend: Python, FastAPI  
OCR: PyMuPDF, Tesseract OCR  
AI Layer: OpenAI API or Gemini API

Innovation  
Unlike traditional ATS systems, this solution uses semantic understanding and explainable AI rather than simple keyword matching.

Impact  
- Reduce screening time by 80–90%  
- Improve shortlist quality  
- Reduce recruiter workload  
- Improve hiring efficiency

Final Pitch  
AI-Powered Recruiter CV Analyzer with Conversational Hiring Assistant is an intelligent recruitment system that automates resume screening using document intelligence and LLM-based reasoning.